MAHATMAGANDHIUNIVERSITY,KOTTAYAM
MGU-UGP (HONOURS) THIRDSEMESTER EXAMINATION

(2024ADMISIONONWARDS)

MG3DSCZGY203– Biological Basis of BehaviourIII
Model Question Paper
Duration:2hrsMaximumMarks:50
Studentsshouldattemptatleastonequestionfromeachcourseoutcometoenhancetheirove rall outcome attainability.

Part A
Fill in the blanks Questions
Answer **All** Questions
Eachquestioncarries**1**mark

1. Sense associated with nasal cavity is known as -----
a)olfaction b) vision c) gustation d) auditory [Remember][CO: 1,2]

2. Which cranial nerve is not associated with gustation?
a)vagus b) olfactory c) glossopharyngeal d) facial [Understand][CO:1, 2]

3. The primary type of cell responsible for detecting odours are called ----receptor neuron
a) olfactory b) auditory c) gustatory d) somatosensory [Remember][CO: 1, 2]

4. Charles Bonnet Syndrome is best characterized by
a) Visual hallucinations in individuals with normal vision and psychiatric illness
b) Auditory hallucinations in elderly individuals with dementia
c) Visual hallucinations in visually impaired individuals with intact cognition
d) Tactile hallucinations following limb amputation [Understand][CO: 3]

5. Which receptor detects temperature changes in the skin?
a) chemoreceptor b) thermoreceptor
c) auditory receptor d) touch receptor [Remember][CO: 2]

6. Which brain structure is primarily involved in the perception of pain?
a) .Hippocampus b) hypothalamus
c) cerebellum d) somatosensory cortex [Remember][CO: 3]

7. Which class of medications is commonly used to stabilize mood in bipolar disorder?
a) Mood stabilisers b) Antidepressants
c)Anxiolytics d)antipsychotics [Remember][CO: 4]

8. Which lifestyle modification is most recommended for managing Poly Cystic Ovarian Syndrome?
a) Regular physical exercise and healthy diet b)Smoking cessation only
c) Increased sugar intake d) Sedentary lifestyle [Apply][CO: 4]

9. Which part of the pituitary gland secretes growth hormone?
a) neurohypophysis b) adenohypophysis
c) hippocampus d) pineal [Remember][CO: 4]

10. The hormone released by the hypothalamus to initiate the stress response is

a) Insulin b) corticotropin releasing hormone

c) Adrenalin d) thyroxine [Remember][CO: 4]

[1x10=10]

Part**B**

ShortAnswerType Questions

Answer **10**Questions

Eachquestioncarries**2**marks

11. Name the different types of special senses. [Remember][CO: 1,2]
12. Explain homeostasis with example. [Understand][CO: 1,2]
13. Comment on visual agnosia. [Understand][CO: 3]
14. Name the nerve that connects the eye to the brain and mention its function. [Understand][CO: 3]
15. What is cochlea and what is its role in hearing? [Understand][CO: 1,2]
16. What are somatosenses? Give two examples. [Understand][CO: 3]
17. Differentiate hyperthyroidism and hypothyroidism. [Understand][CO: 4]
18. What is Premenstrual Dysphoric Disorder (PMDD)? Describe the physiological and psychological impacts of PMDD on daily life. [Apply][CO: 4]
19. What are the main functions of testosterone and estrogen? [Understand][CO: 4]
20. Describe myopia and hyperopia [Understand][CO: 3]
21. What is postpartum depression? Which hormone's sudden drop after childbirth is linked to postpartum depression? [Understand][CO: 4]
22. What is the HPA axis? Name its three components. [Understand][CO: 4]

[2x10=20]

Part**C**

Short EssayTypeQuestions Answer **5** Questions

Eachquestioncarries**4**marks

23. Describe the neural pathway of olfaction. [Understand][CO: 1,2]
24. Elaborate the different types of taste. [Understand][CO: 1,2]
25. Draw a neatly labelled diagram of the human eye and describe the function of each part. [Understand][CO: 3]
26. Describe the nervous connections of the ear. [Understand][CO:1, 2]
27. Describe the anatomical and functional relationship between the hypothalamus and the pituitary gland. [Understand][CO: 4]
28. Explain the role of the adrenal medulla in the ‘fight or flight’ response. [Understand][CO: 4]
29. Give a brief account of the interaction between mental health and endocrine function in anxiety disorders. [Apply][CO: 4]

[4x5=20]